



Figure 5: Virtualisation and cloud computing

environment that has progressed to allow resources anywhere on a network to be dynamically provisioned and consumed, based on application and user requirements. All this would be across a dynamic IT infrastructure, which would be extremely automated, inter-linked and structured to support business procedures, instead of data being isolated in silos.

Typically, organisations advance through various stages of virtualisation. Virtualisation drives agile business solutions, for instance, by resolving specific issues like charge backs, and global concerns such as workload balancing and time to market.

The first stage involves virtualising the ‘low hanging fruit’. Consolidation and disaster recovery strategies are at first targeted to earn returns on capital investments from virtualising programmes that have low business impact. Server virtualisation is also a very popular practice, with many organisations wanting to bring down both CAPEX and OPEX levels.

The second level of virtualisation is when things get confusing, mainly due to the complex design in Level 1. Many CIOs now have a ‘Virtualisation first’ policy, to enjoy the cost benefits. In this second level, companies have begun to use applications, servers, storage, and networks as pools of resources that can be managed in aggregate rather

than isolated silos. Organisations may have experienced unanticipated below-par performance during Level 1 virtualisation, so they will need help to guarantee efficiency of the business critical applications that are virtualised in Level 2 and beyond.

Desktop virtualisation does need a significant change in the infrastructure; so it could be the end of 2012 before we see desktop virtualisation adoption in the millions with enhanced security, manageability and adaptability. The price of desktop virtualisation is a hurdle to adoption, but over the next few years, the price per user will come down continuously.

Many enterprise-level implementations of Level 2 virtualisation store the ‘virtualised’ desktop on a remote server, instead of on the local storage. Thus, when customers work from their local machines, all the applications, procedures, and data used are kept on the hosting server and run centrally. This allows customers using mobile phones or thin clients with very rudimentary hardware specs to run OSs and applications that would normally be beyond their capabilities.

Mobile virtualisation is a technological innovation that allows several OSs or virtual machines to run at the same time on a mobile phone or connected wireless device. It uses a hypervisor to create a secure separation between the hardware and the application that operates on top of it. In 2008, the telecom industry became interested in using the benefits of virtualisation for mobile phones and other gadgets like tablets, net-books and machine-to-machine (M2M) modules. With mobile virtualisation, manufacturing feature-rich phones has become easier through the re-use of applications and hardware, which reduces the length of the development time. One such example is using mobile virtualisation to make low-cost Android smart-phones. Semiconductor companies such as ST-Ericsson have implemented mobile virtualisation as part of their low-cost Android platform strategy.

Comparison of open source virtualisation technology

	KVM	Xen	Linux VServer	OpenVZ
Virtualisation type	Full virtualisation	Para-virtualisation hardware assisted	Container-based virtualisation (OS Level)	Container-based virtualisation (OS Level)
OS Support	Windows and Linux	Windows and Linux	Linux	Linux
Licence	GPL	GPL	GPL	GPL
Guest OS SMP available	Yes, up to 16 VC-PU's per VM	Yes, v4.0.0: up to 128 VCPUs per VM	Yes	Yes
Emulation	Partial	No	No	No
Creator	Qumranet	Xensource	Community Project	Community Project